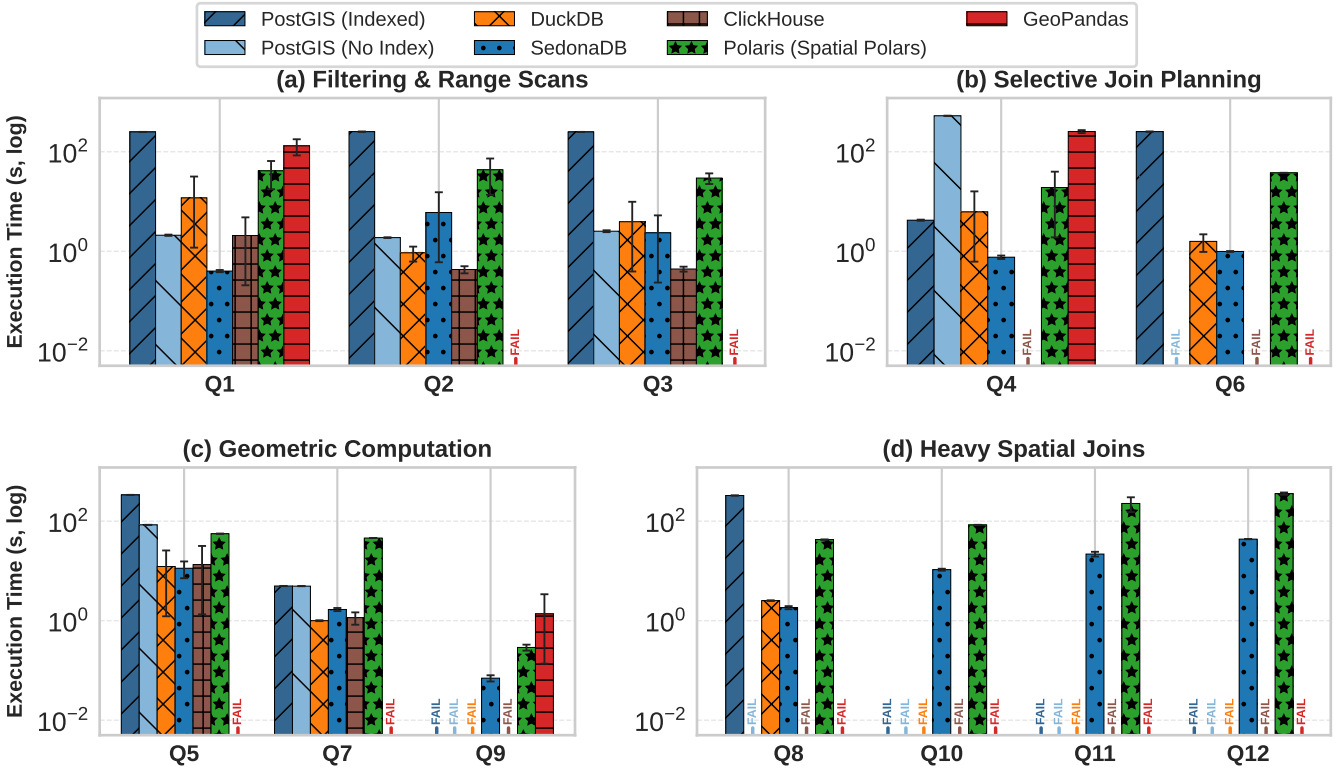
PostGIS (Indexed)
PostGIS (No Index)
DuckDB
SedonaDB
ClickHouse
Polaris (Spatial Polars)
GeoPandas
(a) Filtering & Range Scans
(b) Selective Join Planning
(c) Geometric Computation
(d) Heavy Spatial Joins
Execution Time (s, log)
10^2
10^0
10^{-2}
FAIL
Q1
Q2
Q3
Q4
Q6
Q5
Q7
Q9
Q8
Q10
Q11
Q12